

## HOMEWORK 6

1. Let  $1 \rightarrow H \rightarrow G \rightarrow F \rightarrow 1$  be a short exact sequence of finite groups such that  $F$  is a cyclic group and the integers  $|H|$  and  $|F|$  are relatively prime. Prove that the sequence is split.

2. For every two nonzero integers  $n$  and  $m$  construct an exact sequence

$$0 \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/nm\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \rightarrow 0.$$

For which  $n$  and  $m$  is the sequence split?

3. Let

$$1 \rightarrow H \rightarrow G \rightarrow F \rightarrow 1$$

be a short exact sequence. Prove that if  $F$  is a free group, the sequence is split.

4. Prove that the infinite dihedral group  $D_\infty$  is generated by two elements of order 2.

5. Prove that every group generated by two elements of order 2 is isomorphic to a quotient group of  $D_\infty$ .

6. Let  $X$  be a subset of a group  $G$ . Prove that  $\langle\langle X \rangle\rangle = \langle Y \rangle$ , where  $Y$  is the union of  $gXg^{-1}$  for all  $g \in G$ . (Here  $\langle\langle X \rangle\rangle$  denotes the normal subgroup generated by  $X$ .)

7. Let  $G$  be the group defined by generators  $a, b$  and relations  $w^3 = e$  for all words  $w$  in  $a$  and  $b$ . Show that  $G$  is finite of order 27. (Hint: Show that  $ab$  and  $ba$  commute and the subgroup generated by  $ab$  and  $ba$  is normal. Deduce that  $|G| \leq 27$ .)

8. Prove that if the free groups  $F(X)$  and  $F(Y)$  for finite sets  $X$  and  $Y$  are isomorphic, then  $|X| = |Y|$ .

9. Find a presentation of the quaternion group  $Q_8$  by generators and relations.

10. Determine the order of the group  $\langle a, b \mid a^5, b^4, aba^{-1}b \rangle$ .